

Isolating Novelty in Theoretical Frameworks

A Comparative Analysis of Topologic, Algebraic, and Algorithmic Claims Against Established Literature

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The pursuit of unified frameworks in theoretical physics and mathematics frequently generates expansive corpora that blur the boundaries between ontological repositioning and genuine empirical discovery. Evaluating such works requires a rigorous separation of concepts, distinguishing the recasting of established theorems into novel vocabularies from the introduction of previously undocumented mathematical relationships or physical measurements. The theoretical corpus produced by Dean Kulik in July 2026 presents a sprawling attempt to derive fundamental physical constants, the structural laws of computation, and the algebraic constraints of quantum field theory from a minimal set of continuation axioms.

A systematic cross-referencing of these claims against established scientific literature reveals that substantial portions of the framework consist of standard mathematics and physics dressed in a relational ontology. For instance, the derivation of lattice gauge dynamics from an edge operator is equivalent to Wilson's lattice gauge theory, and the algebraic prohibition of accumulating costs on idempotent semigroups is a known triviality in tropical mathematics. However, when stripped of its pedagogical repackaging, the corpus yields several highly specific, genuinely novel contributions. These include the geometric derivation of the Bailey-Borwein-Plouffe (BBP) formula parameters, the empirical refutation of monotonic entropy consumption in cryptographic diffusion, the statistical deconstruction of the Koide mass relation, and the formalization of the "swept measure" observable. This analysis isolates and details these unique contributions, explicitly articulating their relevance and implications for the broader scientific community.

Topological Derivations in Number Theory: The BBP Equivalence Classes

The Bailey-Borwein-Plouffe (BBP) formula, identified in 1995 via the PSLQ integer relation algorithm, revolutionized experimental mathematics by enabling the extraction of arbitrary hexadecimal digits of π without the computation of preceding digits. Within the established mathematical literature, the parameters of the BBP formula—specifically the base 16, the shell spacing of 8, and the coefficient weight

vector $\{4, -2, -1, -1\}$ —are largely treated as the arbitrary, empirical outputs of lattice reduction searches. While the existence of "zero relations" or "null formulas" for BBP-type series is well-documented, with researchers demonstrating that constants can be represented in multiple overlapping algebraic forms, the geometric origins of the specific parameters have remained untheorized.

The novelty in the analyzed corpus lies not in the discovery of null vectors, but in the formulation of a binary compatibility constraint that mathematically forces the BBP parameters, stripping the formula of its status as an arbitrary empirical artifact. Furthermore, the analysis rigorously identifies the BBP vector as the minimal norm representative of an infinite equivalence class, profoundly reframing the operation of integer relation algorithms.

Geometric Derivation of Radix and Spacing

Standard approaches to BBP formulas rely on the partial fraction expansion of polylogarithmic constants. The alternative framework shifts this to a topological constraint problem. The argument posits that the denominator $1 - x^n$ represents a mathematical kernel whose zero set defines a frame of n -th roots of unity. Under the exponent decomposition $nk + r$, the application of the n -th root of unity acts as a quotient map. The winding count k is annihilated by projection, leaving only the orientation residue r as a surviving geometric angle.

For a digit-extraction formula to function, the upper limit of integration u must simultaneously satisfy two independent conditions. First, it must be a frame-compatible endpoint, meaning it corresponds to the coordinate of one of the frame's marks, such that $u = \cos(\theta)$. Second, it must be a radix-compatible endpoint, meaning the limit raised to the shell spacing must equal the reciprocal of an integer base, such that $u^n = 1/b$.

An exhaustive computational sweep was executed over all regular frames with integer degree measures (up to $n=360$) and all integer bases from 2 to 20. The empirical results reveal that exactly two angles satisfy this joint binary compatibility condition:

Angle	Cosine	Power Representation	Geometric Triangle
45°	0.7071067812	$2^{-1/2}$	45-45-90 (1, 1, $\sqrt{2}$)
60°	0.5000000000	2^{-1}	30-60-90 (1, $\sqrt{3}$, 2)

This geometric lock strictly forces the parameters of the formula. The requirement that $u^n = 1/b$ under the cosine of 45° forces the pairing of base 2^m with a spacing of $n = 2m$. Because 45° requires a minimum of 8 sectors to exist as a named marked direction within a regular frame, and the cosine of 45° requires $n=8$ to satisfy the integration limit condition, the parameters are logically locked at $n=8$, $m=4$, yielding base 16. Base 16 is thus established not as a convenience for binary hardware, but as the absolute minimum mathematical frame in which the target angle of 45° (the angle whose tangent is 1, yielding $\pi/4$) possesses a defined coordinate.

Expanding the Classification: Hexagonal Base-64 Relations

By identifying 60° as the sole alternative binary-compatible angle, the framework predicts the necessary existence of a "hexagonal" family of BBP relations where 6 divides n , utilizing base 2^n . Expanding the frame to $n=6$ and the base to 64, integer relation detection at 60-digit precision confirms this prediction, yielding two previously unexamined BBP-type relations for π .

Relation Target	Frame	Base	Hexagonal Series Denominators over $64k$	Absolute Error
$4\pi\sqrt{3}$	$n=6$	64	$20/(6k+1) + 6/(6k+2) - 1/(6k+3) - 3/(6k+4) - 1/(6k+5)$	4.3×10^{-50}
$4\pi/\sqrt{3}$	$n=6$	64	$12/(6k+1) - 6/(6k+2) - 3/(6k+3) - 3/(6k+4)$	1.1×10^{-50}

These base-64 expansions inherit the tangent of their characteristic frame angle; because the tangent of 60° is $\sqrt{3}$, the resulting constants carry the algebraic residue of $\sqrt{3}$. This structural derivation successfully extends the octal geometric framework into a verified hexagonal regime, proving that the standard BBP formula is not a singular mathematical anomaly but the minimal octal member of a rigorously defined frame-winding classification.

The Null Lattice and Vector Minimality

While the discovery of zero relations in BBP formulas is established, the analysis of the coefficient vector's Jacobian rank provides a novel explanation for the necessity of specific algorithmic discovery tools. Running integer relation detection on the eight shell constants B_r alone, without a target constant, extracts an explicit null vector:

$$-8B_1 + 8B_2 + 4B_3 + 8B_4 + 2B_5 + 2B_6 - 1B_7 + 0B_8 = 0$$

This relation, verified to 4.9×10^{-80} at 80-digit working precision, establishes that the standard BBP constraint matrix has a rank of 1 on 8 unknowns. This leaves a 7-dimensional solution surface where the observable (π) is exactly invariant. Consequently, π admits infinitely many integer representations within the exact same frame.

The analysis proves that neither the derivative screen nor integrality can uniquely select the standard coefficient vector. The sole mathematical selector is minimality. By demonstrating that the gradient of the squared norm constraint ($C_2(w) = \|w\|^2$) is not orthogonal to the null direction ($\nabla C_2 \cdot \text{null} = -104 \neq 0$), minimality is shown to act as a genuine rank-reducing constraint.

Offset (t)	Weight Vector (w+t·null)	Squared Norm ($\ w\ _2^2$)	Equals π
\$-2\$	$[20, -16, -8, -18, -5, -5, 2, 0]$	1098	Yes
\$-1\$	$[12, -8, -4, -10, -3, -3, 1, 0]$	343	Yes
\$0\$	$[4, 0, 0, -2, -1, -1, 0, 0]$ (Standard BBP)	22	Yes
\$1\$	$[-4, 8, 4, 6, 1, 1, -1, 0]$	135	Yes

This mathematically reframes the BBP formula. The standard coefficient vector is not the "correct" solution; it is merely the canonical representative of an equivalence class defined modulo the null lattice. Because the search landscape is exactly flat along the null direction—changing distances along the tangent space produce zero local slope toward the short vector—gradient descent is mathematically blind to the solution. This geometry dictates that the discovery tool must be lattice reduction (such as PSLQ), as it is the only algorithmic class constructed specifically to return the shortest vector in an equivalence class.

Algebraic Information Theory: The Idempotence Separation

In analyzing the constraints that govern computational interfaces, the corpus introduces the "Idempotence Separation" lemma. The formal algebraic statement asserts that no layer whose composition law is idempotent can carry a cost that accumulates homomorphically under that composition. If an operation satisfies $f(x \circ y) = f(x) + f(y)$ and the underlying structure is an idempotent band ($x \circ x = x$), the algebra forces $f(x) = f(x \circ x) = 2f(x)$, meaning $f(x)$ must equal 0.

When cross-referenced against established algebraic literature, this theorem is a trivial property of idempotent semirings, tropical geometry, and ordered algebraic structures. The literature on meet-semilattices routinely utilizes the fact that additive homomorphisms on idempotent structures collapse to zero. The algebraic proof offered in the corpus provides no mathematical novelty, a fact eventually conceded within the corpus's own errata logs, which note that the modular valuation condition on a power set of four elements yields a solution space of dimension 5, while the meet-homomorphism space yields dimension 0.

However, the application of this mathematical triviality to information theory and computational architecture yields a highly novel and measurable physical consequence. The analysis shifts from abstract algebra to the measurement of conditional entropy under pushforward measures, investigating what occurs physically when an idempotent operation is applied twice. Testing all 41 idempotent maps available on a 4-state system, the "crossing price" (defined as $H(X|Y)$) is evaluated.

Measurement on 4-State Systems	Result
Idempotent maps	41 of 256
Maximum drift $\ X(f \circ f) - (X(f) + X_{\text{push}}(f))\ $	0.000×10^3
Idempotents with nonzero second-pass toll	0 of 41
Naive same-measure doubling error	max 2.0000 bits, mean 0.9821 bits

This demonstrates exact "filter saturation." The first application of an idempotent map collapses the source measure onto its image; the second application operates on a measure where it acts injectively. Because the map is the identity on its own image, the conditional entropy of the second pass is zero to machine precision.

By proving that naive, same-measure accounting overcharges for repeated constraints (producing a mean doubling error of nearly 1 bit), the analysis structurally forces a "two-ledger" computational architecture. The system must split accounting into an idempotent admissibility layer (paid once, governed by intersection/meet) and an accumulating execution layer (paid per crossing, governed by sequential monoid composition). This mapping of abstract semigroup limitations directly to thermodynamic and computational costs provides an original and rigorously quantified synthesis, dictating that any hardware or software instruction set that charges twice for the same constraint is misclassifying an execution layer as an admissibility layer.

Swept Measure Observables and the Dimension Lift

Standard physics typically treats observables as intrinsic properties carried by a state—an object possesses mass, area, or charge, which measurement then extracts. The corpus develops an alternative relational ontology: the observable is explicitly the invariant of the orbit that a state traces under its admissible transformations. The template is minimal: a configuration combined with a group of admissible transformations forms an orbit, and the invariant belongs solely to that orbit.

While relativistic mass as an invariant of the Lorentz group and spin magnitude as an invariant of $SU(2)$ are standard concepts in established physics, the extension of this template to the concept of the "swept measure" provides a novel mathematical vocabulary for diagnosing zero-readings in geometry and physics.

The immediate consequence of this framework is that a quantity can read exactly zero without implying absence, provided it is evaluated in the wrong transformation family. The corpus identifies Heron's formula for the area of a triangle as the trivial-group case. Heron's formula computes the measure of the orbit under

the identity group alone. For a collinear configuration of three points, the orbit is simply the configuration itself, correctly returning a measure of zero. The zero is not a failure of the formula; it is the correct measure of the identity orbit.

To demonstrate the generative power of the transformation group over the object, the corpus evaluates the swept Lebesgue measure of a unit segment under various transformation families:

Group Applied to the Unit Segment	Orbit Geometry	Swept Measure	Contains π ?
$SO(2)$ rotation about endpoint	Unit disc	3.141593	Yes
Translation by 1 (perpendicular)	Unit square	1.000000	No
Translation by 2 (perpendicular)	1×2 rectangle	2.000000	No
Shear by 1	Parallelogram	1.000000	No

A control establishes that the constant π does not reside within the unit segment itself. Sweeping the segment via translation or shear yields integer measures. Furthermore, sweeping the $SO(2)$ rotation only partway yields strictly proportional fractions: 0.125π at 45° , 0.5π at 180° , and π exactly at closure (360°). This rigorously reframes π not as a property of a circle, but as the swept measure that a compact one-parameter group assigns to a unit generator exclusively at the point of geometric closure.

To formalize this, the framework introduces the "dimension lift," defined as the dimension of the orbit minus the dimension of the configuration. A dimension lift of 1 (as seen in the rotation of the segment) indicates that the transformation generated the dimension in which the measurement lives. A lift of 0 (as in measuring a pre-existing 3-4-5 right triangle) indicates the dimension was already occupied. This provides a clean integer metric to distinguish dynamically generated geometric content from pre-existing static content.

Relativistic Field Dynamics and the Mass Spectrum

The corpus attempts to build a fundamental physics framework from local graph edge dynamics. A density is assigned to each vertex and a flow to each edge, with the flow updated based on a retained fraction of the old flow minus a coupling to the density difference across the edge. Extending this to matrix-valued quantities forces the introduction of a parallel transporter, yielding a non-abelian covariant difference and generating curvature as the non-commutativity of the edge connection.

Comparing this to established literature, this construction is indistinguishable from standard numerical analysis techniques for discretizing the continuity equation and the telegrapher's equation, combined with

the fundamentals of Kenneth Wilson’s lattice gauge theory. The emergence of the heat equation at zero retention, the wave equation at unit retention, and non-abelian field strengths from matrix commutators are well-known mathematical physics results, offering zero novelty in their derivation.

The Boundary Requirement for Persistent Modes

Where the physics framework achieves originality is in its strict axiomatization of the mass term not as an imported property, but as a mandatory bounded-continuation requirement at zero gradient. The governing constraint is articulated precisely: every admissible state must participate in a defined continuation relation, and for a state to persist, that relation must admit a bounded solution.

By evaluating a perfectly uniform scalar field configuration, spatial curvature drops to zero, disabling all derivative-based evolution. The corpus measures the resulting field dynamics for massless and massive modes:

Configuration	Initial ϕ	Final ϕ	Max Amplitude	Persistence Classification
Massless, zero initial rate	1.0000	1.0000	1.0000	Static (Frozen)
Massless, nonzero initial rate	1.0010	5.0000	5.0000	Runaway (Divergent)
Massive, zero initial rate	0.9984	- 0.9821	1.0002	Bounded Oscillation
Massive, nonzero initial rate	0.9994	- 0.9769	1.0000	Bounded Oscillation

For a massless field, the uniform mode possesses only two branches, neither of which is a persistent physical state. The zero-rate branch is statically frozen, failing the continuation constraint. The nonzero-rate branch is a straight line in time that grows linearly without bound, failing the physical boundedness constraint necessary to prevent infinite energy accumulation.

Only the massive field, which incorporates a term that reads the absolute field amplitude rather than its spatial difference, achieves bounded, persistent oscillation. Simulated at a mass-squared coefficient of 400, the predicted angular frequency is 20.0000, and the measured frequency from zero crossings is 20.0011 (an error of 0.006%). This physically corresponds to the Compton frequency (e.g., 7.7634×10^{20} rad/s for an electron). The analysis proves a rigorous biconditional for second-order relativistic fields: a field admits persistent homogeneous solutions if and only if it carries a mass term. This theoretically explains the

established phenomenological fact that only massive particles possess rest frames, as a system without a spatial gradient requires an internal temporal clock to prevent divergent runaway dynamics.

Statistical Nullification of the Koide Relation

The most impactful contribution to high-energy physics within the corpus is not a derivation, but a rigorous statistical falsification. In the search for the exact coefficients of the mass term, the corpus evaluates the empirical mass ratios of the Standard Model fermions. Standard geometric operator spectra (particle in a box, harmonic oscillator, spherical Laplacian, circular drum) categorically fail to reproduce the lepton mass ratio of 206.768, with the highest standard spectral ratio topping out at 5.269.

The corpus then turns to the Koide relation, an empirical formula noted in 1981 stating that the sum of the charged lepton masses divided by the square of the sum of their square roots approximates $\frac{2}{3}$ to a remarkable relative deviation of 9.233×10^{-6} . The established literature has frequently cited this precision as evidence of undiscovered family symmetries or preon models.

Kulik subjects the Koide relation to severe statistical controls. The function's mathematical boundaries strictly limit its output between $\frac{1}{3}$ and 1 . The value $\frac{2}{3}$ is the exact midpoint of this range. Generating a distribution of two million random mass triples spanning the observed physical ranges reveals a massive pile-up at this midpoint: 2.34% of completely random triples land within 0.01 of the target.

More devastatingly, out of the two million random draws, exactly 24 triples matched the $\frac{2}{3}$ target *better* than the actual physical leptons do. This establishes a baseline occurrence rate of 1.2×10^{-5} (roughly 1 in 83,000) for the precision observed in nature. Because the functional form of the relation was selected retroactively after observing the lepton masses, and the space of simple algebraic functions of three variables is vast, a 1-in-83,000 occurrence rate falls entirely within the expected tail of a random distribution. The corpus successfully demystifies the Koide relation, reclassifying it as a highly predictive constraint that leaves two free parameters, rather than as evidence for an underlying unified geometric theory.

Computational Axiomatics and the Non-Monotonicity of Diffusion

In the domain of computation and grid geometries, the corpus seeks to isolate the structural invariants of discrete numbers and strings when projected into spatial matrices. The analytical instrument is the "coherence functional," a single metric that aggregates block agreement and neighbor agreement across six directional readings (rows, columns, and two diagonal families). Validating the instrument across various substrates (mathematical constants, random bitstreams, substitution-permutation ciphers) demonstrates that it successfully distinguishes structured geometry from random noise without parameter tuning.

The Boundary Budget and Information Channel

Applying the coherence functional to the growth of square grids yields an unconventional conservation law. The interface accounting for a grid of side N is split into cells that agree with their inward neighbor (A) and cells that disagree (B). The total interface budget strictly equals the perimeter: $A + B = 4(N - 1)$, holding without exception across all measured shells from $N=3$ to $N=161$.

Because the active boundary grows linearly while the enclosed interior grows quadratically, the ratio of the boundary to the interior drops from 0.8889 at $N=3$ to 0.0392 at $N=101$. This establishes a "frozen seed" geometric architecture. When a grid grows by wrapping shells, the center cell is set once at $N=1$ and is never rewritten by subsequent growth. The center bit is thus a memoryless, passive sample of the initial fill state.

The corpus introduces the "break rate"—defined as $B / 4(N-1)$ —as the definitive information channel of the grid. A break rate of 0 defines a frozen crystal displaying perfect inheritance, while a break rate of 0.5 defines a formless gas with maximal new information at every shell. The quantification of how new structural rings depart from the history encoded beneath them provides a robust, domain-agnostic metric for analyzing cellular automata and phase transitions.

Falsifying Cryptographic Avalanche Monotonicity

The most consequential algorithmic finding arises when the coherence functional is applied to the internal state variables of a cryptographic hash function. Standard cryptographic theory assumes that the permutation and substitution rounds of a hash function operate as a relentless diffusion mechanism (the avalanche effect), monotonically destroying structural relationships until the state is indistinguishable from random noise.

The overall measurements align with standard theory: across 300 distinct messages, the mean coherence drops from 0.0602 at round 0 to 0.0028 at round 63, a net negative change of -0.0574 representing the consumption of geometric structure. However, analyzing the coherence round-by-round reveals a critical structural anomaly: the destruction of structure is definitively not monotonic.

Out of 300 execution traces, exactly zero were strictly monotonic. Furthermore, applying an 8-window smoothing function still yielded zero monotonic traces. The individual diffusion rounds reliably "wobble," with a single round frequently increasing local geometric agreement before subsequent rounds destroy it. This definitively withdraws the long-standing assumption that naive entropy or coherence acts as a strict Lyapunov function for cryptographic diffusion layers. The ordering of chaos is guaranteed only at the endpoints, meaning that intermediate cryptographic rounds temporarily reconstruct latent symmetries. This non-monotonicity presents a theoretical vulnerability, suggesting that if specific geometric projections matching these temporary coherence spikes could be isolated, they could be exploited in advanced differential cryptanalysis.

The Axiomatics of Universal Receivers

At the lowest level of computational axiomatics, the corpus evaluates the fundamental constraint C_{11} (distinction must continue without halting). Evaluating all four possible mappings on a minimal two-state system reveals that fixed points fail the continuation constraint. Exhaustive mapping leaves exactly one survivor: the swap operation. This demonstrates that computational reversibility at the base layer is not an independent physical axiom (as often treated in Landauer-Bennett reversible computing paradigms) but is rather strictly forced by the continuation requirement on a minimal binary alphabet.

This leads to a rigorous examination of system receivers and admissibility. The corpus proves geometrically that a finite receiver cannot be both universally admissible and fully discriminating. A complete deterministic finite automaton (DFA) forced to map k^L strings into N states where $k^L > N$ must invoke the pigeonhole principle and merge paths, permanently destroying historical information. The only receiver capable of universal continuation without state merging is the free monoid, which scales its state infinitely with its history. This provides a formal mathematical proof for why perfectly lossless physical systems must inherently encode their entire interaction history into an ever-expanding phase space.

Synthesis

While the Kulik corpus dedicates extensive volume to the ontological reframing of established mathematics and physics—particularly in its treatments of lattice gauge theory and idempotent algebraic structures—rigorous comparative analysis isolates several highly specific, genuinely novel contributions.

The geometric lock of the BBP formula parameters successfully derives the hexadecimal radix and shell spacing from topological constraints, identifying the coefficient vector as merely the minimal representative of an infinite null lattice equivalence class, and extending the framework to verified base-64 hexagonal relations. In physics, the formulation of rest mass as the mandatory bounded-continuation term at zero gradient is accompanied by a severe statistical demystification of the Koide relation, demonstrating its precision to be a routine artifact of random numerical distributions. Finally, the application of coherence functionals to cryptographic grids proves that diffusion in hash rounds is non-monotonic, exposing intermediate structural reconstructions in processes previously assumed to be strictly entropic. These isolated findings present robust, verifiable advancements across number theory, physical statistics, and algorithmic geometry.

Works cited

1. Series that converge to π quickly - Math Stack Exchange, <https://math.stackexchange.com/questions/14113/series-that-converge-to-pi-quickly>
2. BBP-type formulas -- an elementary approach - ResearchGate, https://www.researchgate.net/publication/362728285_BBP-type_formulas_-_an_elementary_approach
3. Kernel_Frame_Winding_Residue.docx
4. Kernel_Frame_Winding.docx
5. The_Angle_Register.docx
6. The_Idempotence_Separation.docx
7. On the Additive and Multiplicative Structure of Semirings, <http://www.researchmathsci.org/apamart/apam-v3n1-8.pdf>
8. Semiring and semimodule issues in MV-algebras - arXiv, <https://arxiv.org/pdf/1002.4013>
9. Idempotent analysis and continuous semilattices - LSU Scholarly Repository, https://repository.lsu.edu/cgi/viewcontent.cgi?article=1619&context=mathematics_pubs

10. An Introduction to Idempotency - Inria, <https://people.bordeaux.inria.fr/pierre.delmoral/HPL-BRIMS-g6-24.pdf>
11. A Survey of Partially Ordered Algebras Peter Jipsen and Bianca Newell - Chapman University, <https://math.chapman.edu/~jipsen/Survey-of-po-algebras-DRAFT.pdf>
12. The_Observable_Is_The_Invariant_Of_The_Orbit.docx
13. An Introduction to Spinors and Geometry with Applications in Physics - rexresearch1, <https://www.rexresearch1.com/SpinorsLibrary/IntroductionSpinorsGeometryApplicationsPhysicsBenn.pdf>
14. Progress in Physics, Vol. III/2013 - Academia.edu, https://www.academia.edu/34889646/Progress_in_Physics_Vol_III_2013
15. 2013 Volume 3, <https://www.ptep-online.com/files/complete/pip-2013-03.pdf>
16. One_Edge_Operator.docx
17. The_Clock_At_Zero_Gradient.docx
18. The_Coherence_Lattice.docx
19. The_Co_Audit_Trail.docx
20. converted-document.docx